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EE 632

Assignment 1

1. A Gaussian random vector $\mathbf{X}$ is distributed as $\mathcal{N}(\boldsymbol{\mu}, \mathbf{K})$, where

$$\boldsymbol{\mu} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \qquad \mathbf{K} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

From $\mathbf{X}$, obtain a white random vector $\mathbf{Y}$ with covaraince matrix $\mathbf{I}_2$ by eigen decomposition such that $\mathbf{Y} = \mathbf{AX}$.

(a) Find $\mathbf{A}$.

(b) Find the probability density function (PDF) and characteristic function (CF) of $\mathbf{Y}^T\mathbf{Y}$.

Sol^n. (a) Given $\bar{X} \sim \mathcal{N}(\bar{\mu}, \bar{K})$

$$\bar{\mu} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \bar{K} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

Eigenvalues of $\bar{K}$:

$$\begin{vmatrix} 3-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = 0$$

$$(3-\lambda)^2 - 1 = 0$$

$$\lambda^2 - 6\lambda + 8 = 0$$

$$(\lambda-2)(\lambda-4) = 0$$

$$\Rightarrow \lambda_1 = 2, \quad \lambda_2 = 4$$

Let $\bar{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$ be an eigenvector for eigenvalue 2

$$\bar{K}\bar{x} = 2\bar{x}$$

$$\Rightarrow \quad 3x_1 + x_2 = 2x_1 \quad - (A)$$

$$\text{and,} \quad x_1 + 3x_2 = 2x_2 \quad - (B)$$

From (A) and (B)

$$x_1 + x_2 = 0$$

Choosing $e_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$

Let $\bar{y} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$ be an eigenvector for eigenvalue 4

$$\bar{K}\bar{y} = 4\bar{y}$$

$$\Rightarrow \quad 3y_1 + y_2 = 4y_1 \quad - \text{ⓒ}$$

$$\text{and,} \quad y_1 + 3y_2 = 4y_2 \quad - \text{ⓓ}$$

From ⓒ and ⓓ,

$$y_1 = y_2$$

Choosing $e_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$

$$\therefore \bar{E} = \begin{bmatrix} e_1 & e_2 \end{bmatrix}$$

$$\bar{E} = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

Now $\bar{E}^{-1} = \bar{E}^{T} = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$

$$\bar{\Lambda}^{-\frac{1}{2}} = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 \\ 0 & \frac{1}{2} \end{bmatrix}$$

$$\bar{B} = \bar{E}\bar{\Lambda}^{\frac{1}{2}}$$

$$\bar{B}^{-1} = \bar{\Lambda}^{-\frac{1}{2}} \bar{E}^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} \end{bmatrix}$$

Hence, if $\bar{X} = \begin{bmatrix} X_1 \\ X_2 \end{bmatrix} \sim \mathcal{N}(\bar{\mu}, \bar{K})$

then $\bar{Y} = \bar{B}^{-1}\bar{X} \sim \mathcal{N}(\bar{B}^{-1}\bar{\mu}, \bar{I}_2)$

$$\bar{Y} = \begin{bmatrix} \frac{X_1 - X_2}{2} \\ \frac{X_1 + X_2}{2\sqrt{2}} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} -\frac{1}{2} \\ \frac{3}{2\sqrt{2}} \end{bmatrix}, \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)$$

Therefore $\bar{A} = \bar{B}^{-1} = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} \\ \frac{1}{2\sqrt{2}} & \frac{1}{2\sqrt{2}} \end{bmatrix}$

(b) Let $Z = \bar{Y}^T \bar{Y}$ and $\bar{Y} = \begin{bmatrix} Y_1 \\ Y_2 \end{bmatrix}$ where $Y_1 \sim \mathcal{N}(-\frac{1}{2}, 1)$ and $Y_2 \sim \mathcal{N}(\frac{3}{2\sqrt{2}}, 1)$
(Y_1 and Y_2 are independent)

$\Rightarrow \quad Z = Y_1^2 + Y_2^2$

Hence, Z follows the non-central χ^2 distribution with $n=2$ degrees of freedom.

$$s = \sqrt{\frac{1}{4} + \frac{9}{8}} = \frac{1}{2}\sqrt{\frac{11}{2}}$$

PDF of Z is then,

$$f_Z(z) = \frac{1}{2\sigma^2}\left(\frac{z}{s^2}\right)^{\frac{n-2}{4}} e^{-\left(\frac{s^2+z}{2\sigma^2}\right)} I_{\frac{n}{2}-1}\left(\frac{s}{\sigma^2}\sqrt{z}\right), \quad z > 0$$

Putting $\sigma^2 = 1, n=2$ and $s = \frac{1}{2}\sqrt{\frac{11}{2}}$

$$f_Z(z) = \frac{1}{2} e^{-\left(\frac{11+8z}{16}\right)} I_0\left(\frac{1}{2}\sqrt{\frac{11z}{2}}\right), \quad z > 0$$

CF of Z, $\varphi_Z(j\omega) = \left(\frac{1}{1-2j\omega\sigma^2}\right)^{\frac{n}{2}} e^{\frac{j\omega s^2}{1-2j\omega\sigma^2}}$

Putting $\sigma^2 = 1, n=2$ and $s = \frac{1}{2}\sqrt{\frac{11}{2}}$

$$\varphi_Z(j\omega) = \left(\frac{1}{1-2j\omega}\right) e^{\frac{j\omega}{1-2j\omega}\left(\frac{11}{8}\right)}$$

2. A zero-mean complex circular Gaussian random vector $\mathbf{Z} = [Z_1 \;\; Z_2]^T$ has the PDF

$$f_{\mathbf{Z}}(\mathbf{z}) = f_{Z_1,Z_2}(z_1, z_2) = \frac{1}{11\pi^2} \exp\left\{-\frac{1}{11}\left(8|z_1|^2 + 2|z_2|^2 + 2\mathrm{Re}\left[(-1+2j)z_1^* z_2\right]\right)\right\},$$

where $z_1, z_2 \in \mathbb{C}$, $\mathbb{C}$ denotes the set of complex numbers, $j = \sqrt{-1}$, and $(\cdot)^*$ is the complex conjugate.

(a) Find the joint CFs $\Psi_{X_1,Y_2}(jw_1, jw_2)$ and $\Psi_{Y_1,X_2}(jw_1, jw_2)$.

(b) Find the joint PDFs $f_{X_1,X_2}(x_1, x_2)$ and $f_{X_1,Y_1}(x_1, y_1)$.

(c) If $\mathbf{U} = \mathbf{X} + \alpha\mathbf{Y} + \begin{bmatrix}1\\1\end{bmatrix}$ and $\mathbf{V} = \mathbf{X} + \beta\mathbf{Y} + \begin{bmatrix}2\\2\end{bmatrix}$, where α and β are real constants, and $\alpha > 0$. Find the values of α and β for which $\mathbf{W} = \mathbf{U} + j\mathbf{V}$ is circular.

Sol^n. We know, when $\bar{Z}$ is a complex Gaussian random vector with zero mean

$$f_{\bar{Z}}(\bar{z}) = \frac{1}{\pi^L |\bar{K}_{\bar{Z}}|} \exp\left\{-\bar{z}^H \bar{K}_{\bar{Z}}^{-1} \bar{z}\right\}$$

From the given value we get,

$$-\bar{z}^H \bar{K}_{\bar{Z}}^{-1} \bar{z} = -\frac{1}{11}\left(8|z_1|^2 + 2|z_2|^2 + 2\mathrm{Re}\left[(-1+2j)z_1^* z_2\right]\right)$$

$$= -\frac{1}{11}\left(8|z_1|^2 + 2|z_2|^2 + (-1+2j)z_1^* z_2 + (-1-2j)z_1 z_2^*\right)$$

$$\Rightarrow \quad \bar{K}_{\bar{Z}}^{-1} = \frac{1}{11}\begin{bmatrix} 8 & -1+2j \\ -1-2j & 2 \end{bmatrix}$$

Taking inverse,

$$\bar{K}_{\bar{Z}} = 11 \cdot \frac{1}{11}\begin{bmatrix} 2 & 1-2j \\ 1+2j & 8 \end{bmatrix} = \begin{bmatrix} 2 & 1-2j \\ 1+2j & 8 \end{bmatrix}$$

Since $\bar{Z}$ is circular $\tilde{K}_{\bar{Z}} = 0_{2\times 2}$,

$$\bar{K}_{\bar{X}\bar{X}} = \bar{K}_{\bar{Y}\bar{Y}} = \frac{1}{2}\mathrm{Re}\{\bar{K}_{\bar{Z}}\} = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 4 \end{bmatrix}$$

$$\bar{K}_{\bar{X}\bar{Y}} = \frac{1}{2}\,\mathrm{Im}\{\bar{K}_{\bar{Z}}\} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = -\bar{K}_{\bar{Y}\bar{X}}$$

Also $\mathbb{E}[\bar{Z}] = \mathbb{E}[\bar{X} + j\bar{Y}] = \mathbb{E}[\bar{X}] + j\,\mathbb{E}[\bar{Y}] = \bar{0} + j\bar{0}$

$\Rightarrow \mathbb{E}[\bar{X}] = \bar{0}$ and $\mathbb{E}[\bar{Y}] = \bar{0}$

(a) For zero mean real random vectors the CF is

$$\varphi_{\bar{V}}(j\bar{\omega}) = \exp\{-\tfrac{1}{2}\,\bar{\omega}^T \bar{K}\,\bar{\omega}\}$$

The covariance matrix of $[X_1 \;\; Y_2]^T$

$$\bar{K}_1 = \begin{bmatrix} 1 & 1 \\ -1 & 4 \end{bmatrix} \qquad (\text{from } \bar{K}_{\bar{X}\bar{X}},\, \bar{K}_{\bar{Y}\bar{Y}} \text{ and } \bar{K}_{\bar{X}\bar{Y}})$$

$$\Rightarrow \text{CF},\; \varphi_{X_1, Y_2}(j\omega_1, j\omega_2) = \exp\left\{-\frac{1}{2}\left(\omega_1^2 + 4\omega_2^2 - 2\omega_1\omega_2\right)\right\}$$

The covariance matrix of $[Y_1 \;\; X_2]^T$

$$\bar{K}_2 = \begin{bmatrix} 1 & 1 \\ 1 & 4 \end{bmatrix} \qquad (\text{from } \bar{K}_{\bar{X}\bar{X}},\, \bar{K}_{\bar{Y}\bar{Y}} \text{ and } \bar{K}_{\bar{X}\bar{Y}})$$

$$\Rightarrow \text{CF},\; \varphi_{X_1, Y_2}(j\omega_1, j\omega_2) = \exp\left\{-\frac{1}{2}\left(\omega_1^2 + 4\omega_2^2 + 2\omega_1\omega_2\right)\right\}$$

(b) For zero mean real random vectors the PDF is

$$f_{\bar{V}}(\bar{v}) = \frac{1}{(2\pi)^{L/2}\,|\bar{K}|^{1/2}} \exp\left\{-\frac{1}{2}\,\bar{v}^T K^{-1} \bar{v}\right\}$$

The covariance matrix of $[X_1 \;\; X_2]^T$

$$\bar{K}_3 = \bar{K}_{\bar{X}\bar{X}} = \begin{bmatrix} 1 & \frac{1}{2} \\ \frac{1}{2} & 4 \end{bmatrix}$$

$$|\bar{K}_3| = 4 - \frac{1}{4} = \frac{15}{4} \quad \text{and} \quad \bar{K}_3^{-1} = \frac{4}{15}\begin{bmatrix} 4 & -\frac{1}{2} \\ -\frac{1}{2} & 1 \end{bmatrix}$$

$$\Rightarrow \text{PDF}, \; f_{X_1, X_2}(x_1, x_2) = \frac{1}{\pi\sqrt{15}} \exp\left\{-\frac{2}{15}\left(4x_1^2 + x_2^2 - x_1 x_2\right)\right\}$$

The covariance matrix of $[X_1 \; Y_1]^T$

$$\bar{K}_4 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = \bar{I}_2 \qquad (\text{from } \bar{K}_{\bar{X}\bar{Y}})$$

$$\Rightarrow \text{PDF}, \; f_{X_1, Y_1}(x_1, y_1) = \frac{1}{2\pi} \exp\left\{-\frac{1}{2}\left(x_1^2 + y_1^2\right)\right\}$$

(c) Given $\bar{U} = \bar{X} + \alpha\bar{Y} + \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\bar{V} = \bar{X} + \beta\bar{Y} + \begin{bmatrix} 2 \\ 2 \end{bmatrix}$

$$\mathbb{E}[\bar{U}] = \mathbb{E}[\bar{X}] + \alpha\,\mathbb{E}[\bar{Y}] + \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\mathbb{E}[\bar{V}] = \mathbb{E}[\bar{X}] + \beta\,\mathbb{E}[\bar{Y}] + \begin{bmatrix} 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$$

Now

$$\bar{W} = \bar{U} + j\bar{V}$$

$$\Rightarrow \mathbb{E}[\bar{W}] = \mathbb{E}[\bar{U}] + j\,\mathbb{E}[\bar{V}] = \begin{bmatrix} 1+2j \\ 1+2j \end{bmatrix}$$

$$\bar{W} = \bar{U} + j\bar{V} = \bar{X} + \alpha\bar{Y} + \begin{bmatrix} 1 \\ 1 \end{bmatrix} + j\left(\bar{X} + \beta\bar{Y} + \begin{bmatrix} 2 \\ 2 \end{bmatrix}\right)$$

$$= (1+j)\bar{X} + (\alpha + j\beta)\bar{Y} + \begin{bmatrix} 1+2j \\ 1+2j \end{bmatrix}$$

For $\bar{W}$ to be circular, $\tilde{K}_{\bar{w}} = \bar{0}$

$$\Rightarrow \mathbb{E}\left[(\bar{W} - \mathbb{E}[\bar{W}]) (\bar{W} - \mathbb{E}[\bar{W}])^T \right] = \bar{0}$$

$$\mathbb{E}\left[((1+j)\bar{X} + (\alpha + j\beta)\bar{Y}) ((1+j)\bar{X} + (\alpha + j\beta)Y)^T \right] = \bar{0}$$

$$(1+j)^2 \mathbb{E}[\bar{X}\bar{X}^T] + (\alpha+j\beta)^2 \mathbb{E}[\bar{Y}\bar{Y}^T] + (1+j)(\alpha+j\beta)\mathbb{E}[\bar{X}\bar{Y}^T] + (1+j)(\alpha+j\beta)\mathbb{E}[\bar{Y}\bar{X}^T] = \bar{0}$$

$$(1+j)^2 \bar{K}_{\bar{X}\bar{X}} + (\alpha+j\beta)^2 \bar{K}_{\bar{Y}\bar{Y}} + (1+j)(\alpha+j\beta)\bar{K}_{\bar{X}\bar{Y}} + (1+j)(\alpha+j\beta)\bar{K}_{\bar{Y}\bar{X}} = \bar{0}$$

Using $\bar{K}_{XY} = -\bar{K}_{\bar{Y}\bar{X}}$ and $\bar{K}_{\bar{X}\bar{X}} = \bar{K}_{\bar{Y}\bar{Y}}$

$$(2j + \alpha^2 - \beta^2 + 2j\alpha\beta)\bar{K}_{\bar{X}\bar{X}} = \bar{0}$$

$\Rightarrow \alpha^2 - \beta^2 = 0 \quad \Rightarrow \alpha = |\beta|$

and $2 + 2\alpha\beta = 0 \quad \Rightarrow \alpha\beta = -1$

Given $\alpha > 0$

$\Rightarrow \alpha = 1$ and $\beta = -1$